

# Evaluating axillary lymph node metastasis risks in breast cancer patients via Semi-ALNP: a multicenter study



Limeng Qu,<sup>a,b,o</sup> Jinfeng Zhu,<sup>a,b,o</sup> Xilong Mei,<sup>c</sup> Zixi Yi,<sup>d</sup> Na Luo,<sup>e</sup> Songlin Yuan,<sup>e</sup> Xuan Liu,<sup>f</sup> Mingwen Liu,<sup>f</sup> Haiqing Xie,<sup>g</sup> Xiongqiang Hu,<sup>g</sup> Liangrui Pan,<sup>h</sup> Qingchun Liang,<sup>i</sup> Yanhui Li,<sup>c</sup> Qiongyan Zou,<sup>a,b</sup> Qin Zhou,<sup>a,b</sup> Danhua Zhang,<sup>a,b</sup> Meirong Zhou,<sup>a,b</sup> Lei Pei,<sup>a,b</sup> Ke Qian,<sup>a,b</sup> Qian Long,<sup>a,b</sup> Qitong Chen,<sup>a,b</sup> Xi Chen,<sup>a,b</sup> Jennifer K. Plichta,<sup>j,k,l</sup> Qingyao Shang,<sup>m</sup> Meishuo Ouyang,<sup>n</sup> Jiachi Xu,<sup>a,b,\*\*</sup> and Wenjun Yi<sup>a,b,\*</sup>



<sup>a</sup>Department of General Surgery, The Second Xiangya Hospital, Central South University, Changsha, Hunan, 410011, China

<sup>b</sup>Clinical Research Center For Breast Disease In Hunan Province, Changsha, Hunan, 410011, China

<sup>c</sup>Department of Radiology, The Second Xiangya Hospital of Central South University, Changsha, Hunan, 410011, China

<sup>d</sup>Mathematics, Central South University, Changsha, Hunan, 410083, China

<sup>e</sup>Breast and Thyroid Surgery, The First People's Hospital of Changde City, Changde, Hunan, 415000, China

<sup>f</sup>Breast and Thyroid Surgery, The First People's Hospital of Xiangtan City, Xiangtan, Hunan, 411100, China

<sup>g</sup>Breast and Thyroid Surgery, The First People's Hospital of Chenzhou City, Chenzhou, Hunan, 423000, China

<sup>h</sup>College of Computer Science and Electronic Engineering, Hunan University, Changsha, Hunan, 410083, China

<sup>i</sup>Department of Pathology, The Second Xiangya Hospital, Central South University, Changsha, Hunan, 410011, China

<sup>j</sup>Duke Cancer Institute, Duke University Medical Center, Durham, NC, USA

<sup>k</sup>Department of Population Health Sciences, Duke University School of Medicine, Durham, NC, USA

<sup>l</sup>Department of Surgery, Duke University School of Medicine, Durham, NC, USA

<sup>m</sup>National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Chaoyang, Beijing, 100021, China

<sup>n</sup>Department of Surgery, Duke University School of Medicine, Duke University, Durham, NC, USA

## Summary

**Background** Accurately evaluating axillary lymph nodes (ALNs) is essential for guiding both staging and treatment strategies in breast cancer (BC) patients. Currently, traditional pathological staging methods still rely on invasive biopsies or surgeries. This study aimed to construct, evaluate, and validate a semisupervised classifier utilizing radiomic and machine learning (ML) techniques to noninvasively identify axillary nodal disease.

**Methods** Data from 4191 ALNs in 494 patients with invasive BC were retrospectively analyzed at the Second Xiangya Hospital of Central South University between January 31, 2016, and July 31, 2024, including a labeled cohort (214 patients, 1769 ALNs, divided into ultra-low and ultra-high risk groups) and an unlabeled cohort (280 patients, 2422 ALNs). Regions of interest (ROIs) were segmented, and CT radiomic features were extracted. 11 supervised learning models were built on the basis of labeled ALNs, and pseudolabels (low-risk, high-risk groups) were assigned to unlabeled ALNs. Seven ML algorithms developed semisupervised multiclassifiers on the basis of the predicted probabilities for 4191 ALNs. For multicenter validation, additional data were collected from the First People's Hospital of Chenzhou City, the First People's Hospital of Changde City, and the First People's Hospital of Xiangtan City. The best-performing multiclassifier was evaluated in two independent multicenter cohorts: 212 clinically node-positive (cN+) patients who underwent core needle biopsy (CNB) or fine needle aspiration (FNA), and 450 clinically node-negative (cN0) patients. The research was registered at [www.isrctn.com](http://www.isrctn.com) with registration number ISRCTN54288903.

**Findings** The supervised multilayer perceptron (MLP) model, built from labeled ALNs, exhibited excellent classification performance, with an area under the curve (AUC) of 0.959 (95% CI: 0.937–0.981), a sensitivity of 0.899, and a specificity of 0.932. Pseudolabels for the unlabeled ALNs were generated via this model, and the semisupervised MLP multiclassifier (Semi-ALNP) was constructed by combining the labeled and unlabeled data. The AUCs for predicting nodal metastases were 0.906 (95% CI: 0.894–0.917), 0.936 (95% CI: 0.928–0.945), 0.948 (95% CI: 0.940–0.956), and 0.955 (95% CI: 0.946–0.965) for the ultra-low risk, low-risk, high-risk, and ultra-high risk groups, respectively. Validation in both the biopsy and cN0 cohorts revealed strong diagnostic performance: in the biopsy

eClinicalMedicine  
2025;85: 103311

Published Online 24 June

2025

<https://doi.org/10.1016/j.eclinm.2025.103311>

\*Corresponding author. Department of General Surgery, The Second Xiangya Hospital of Central South University, No. 139, Renmin Central Road, Changsha, 410011, China.

\*\*Corresponding author. Department of General Surgery, The Second Xiangya Hospital of Central South University, No. 139, Renmin Central Road, Changsha, 410011, China.

E-mail addresses: [yijwenjun@csu.edu.cn](mailto:yijwenjun@csu.edu.cn) (W. Yi), [xujiachi@csu.edu.cn](mailto:xujiachi@csu.edu.cn) (J. Xu).

<sup>o</sup>These authors have contributed equally to this work.

cohort, the model achieved a false negative rate (FNR) of 1.21%, a false positive rate (FPR) of 14.89%, a sensitivity of 98.79%, and a specificity of 85.11%; in the cN0 cohort, the FNR was 8.33%, the FPR was 9.94%, the sensitivity was 91.67%, and the specificity was 90.06%.

**Interpretation** Semi-ALNP, which is based on the MLP algorithm, has high accuracy in assessing the statuses of ALNs across all types of BC patients. It is particularly effective for identifying high-risk patients with ALN metastasis, which can help guide personalized treatment decisions. Future prospective studies are planned to further validate the clinical utility of this approach in real-world settings.

**Funding** This study was funded by the Science and Technology Innovation Program of Hunan Province (Grant No. 2021SK2026) and the Innovation Platform and Talent Plan of Hunan Province (2023SK4019). Funding sources were not involved in the study design, data collection, analysis and interpretation, writing of the report, or decision to submit the article for publication.

**Copyright** © 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

**Keywords:** Breast cancer; Axillary lymph node; CT radiomics; Machine learning; Semi-supervised classifier

#### Research in context

##### Evidence before this study

We systematically searched PubMed and Web of Science through April 1, 2025, using the terms “breast cancer,” “axillary lymph node,” “radiomics,” and “machine learning.” Existing radiomics-based models for ALN assessment were primarily supervised, focused on binary patient-level classification, and often inferred nodal status indirectly from primary tumor features. No prior studies have successfully applied a semi-supervised approach integrating labeled and unlabeled data for direct, multiclass lymph node-level prediction.

##### Added value of this study

This study is the first to develop and externally validate a semisupervised, multiclass radiomic classifier (Semi-ALNP) that stratifies ALN metastasis risk into four distinct categories—ultra-low, low, high, and ultra-high—by

leveraging both labeled and unlabeled data from 4191 ALNs in 494 breast cancer patients. Through the integration of supervised learning with pseudolabeling techniques, the model demonstrated high predictive accuracy and strong generalizability. Its clinical utility was further confirmed in two independent multicenter validation cohorts—One cohort consisted of clinically node-positive (cN+) patients who had undergone axillary biopsy, while the other cohort was composed of clinically node-negative (cN0) patients.

##### Implications of all the available evidence

The Semi-ALNP model demonstrates the potential of semisupervised radiomics to noninvasively and accurately predict individual lymph node status. With future testing this approach may reduce reliance on invasive procedures such as core needle biopsy or fine needle aspiration while enabling timely identification of high-risk patients.

## Introduction

Axillary lymph node metastasis (ALNM) is the most frequent site of metastasis in patients with breast cancer (BC) and serves as a critical independent prognostic factor.<sup>1,2</sup> Accurately evaluating axillary lymph nodes (ALNs) is essential for guiding both the staging and treatment strategies applied to BC patients.<sup>1,3,4</sup> However, clinical ALN assessments often lack the precision needed for definitive staging, leading to continued reliance on invasive procedures such as needle biopsy and/or axillary surgery.<sup>5–8</sup> Core needle biopsy (CNB) and fine needle aspiration (FNA) are established axillary evaluation methods and may yield positive results that serve as important criteria for determining treatment. Despite their utility, these percutaneous techniques are highly dependent on the type of imaging, ALN size, ALN location, and operator expertise, with potentially

suboptimal performance leading to delayed or missed treatment opportunities. For clinically ALN-negative (cN0) patients, sentinel lymph node biopsy (SLNB) reduces the risk of complications associated with axillary lymph node dissection (ALND), but up to 20% of cN0 patients still harbor ALNMs, which may impact treatment outcomes.<sup>8,9</sup> This underscores the need for noninvasive, more precise diagnostic techniques.<sup>10</sup>

Radiomics, which is an innovative technique for quantitatively analyzing medical imaging data, has substantially increased the accuracy of disease diagnoses.<sup>11–13</sup> However, the extensive and complex nature of these radiomic features poses challenges in manual classification scenarios. Machine learning (ML) algorithms have emerged as powerful tools for assisting with disease diagnosis and treatment by simulating human behaviors to achieve intelligent learning and

data processing.<sup>14–16</sup> Recently, radiology-based ML prediction methods have made significant progress in terms of identifying ALNs.<sup>17–21</sup> Shawn Sun et al. developed a convolutional neural network (CNN)-based ALNM prediction model using ultrasound images of primary BC patients. The model achieved an accuracy of 72.6%, with a sensitivity of 65.5% and a specificity of 78.9%.<sup>22</sup> Bong-Il Song successfully developed a radiomics model using positron emission tomography (PET)/CT images to predict ALNM, and the model achieved a sensitivity of 90.9%, a specificity of 71.4%, and an accuracy of 80% in a test cohort.<sup>23</sup> Tan et al. extracted radiomic features from fat-suppressed T2-weighted images of breast MR images. They developed a linear regression prediction model incorporating clinicopathologic factors and radiomic features. The model achieved a sensitivity of 60.87%, a specificity of 89.19%, an accuracy of 78.33%, and an area under the curve (AUC) of 0.810.<sup>24</sup> Notably, these recent methods have focused mainly on ALNM prediction on the basis of primary BC lesions, and few studies have focused on ALNM prediction on the basis of features derived directly from the lymph nodes themselves.<sup>24–28</sup> Integrating radiomic features of ALNs with ML algorithms offers significant potential for enhancing ALN evaluation and aiding treatment decisions in BC patients.<sup>29–31</sup>

In a previous study, we constructed an unsupervised prediction model based on unlabeled ALNs, which achieved good diagnostic efficacy, with an AUC of 0.847 (0.825–0.869), a sensitivity of 80.70%, and a specificity of 88.84%.<sup>32</sup> Although unsupervised models do not depend on labeling information and simplify the sample acquisition process, their efficacy is somewhat diminished by training exclusively on unlabeled data. In contrast, semisupervised prediction models can leverage both labeled and unlabeled data for training.<sup>33,34</sup> This approach not only facilitates the discovery of potential patterns or structures through unlabeled data, but also enhances the overall performance of the constructed model by effectively utilizing the available labeled data.<sup>35,36</sup>

Therefore, this study describes the development of a novel semisupervised classifier for ALN evaluation in BC, built on a large-scale dataset of 4191 node-level CT segmentations. It combines rigorous feature selection with a data-driven pseudo-labeling strategy, employs patient-level GroupKFold cross-validation, and is externally validated in two independent cohorts. This integrated approach substantially enhances ALN staging accuracy and reliably identifies patients at high risk of nodal metastasis, thereby informing personalized treatment decisions (Fig. 1).

## Methods

### Patient selection

In this retrospective study, data were collected from 494 female patients with unilateral invasive BC who

underwent preoperative high-resolution thin-section chest-enhanced CT and ALND at the discretion of the treating oncology team at the Department of Breast Surgery, Second Xiangya Hospital, Central South University, between January 31, 2016, and July 31, 2024. These patients had complete clinicopathological data, no evidence of distant metastases, and had not received neoadjuvant therapy (NAT) (Fig. 2).

Clinicopathological data were retrieved from the electronic medical records system of the hospital. The recorded clinicopathological factors included age, menopausal status, pathological TNM stage, pathological examination date, tumor type, pathological grade, estrogen receptor (ER) status, progesterone receptor (PR) status, human epidermal growth factor receptor 2 (HER2) status, tumor proliferation index (Ki-67), subtypes, number of ALNs with metastases, and total number of ALNs removed. The CT-related information included the hospitalization number, CT number, and CT examination date of each patient.

For multicenter validation, medical records were collected from ALN biopsy and cN0 BC patients treated at four hospitals between January 1, 2023, and July 31, 2024: the Second Xiangya Hospital, the First People's Hospital of Chenzhou City, the First People's Hospital of Changde City, and the First People's Hospital of Xiangtan City.

For the ALN biopsy cohort, which included 212 patients, clinically positive ALNs (cN+) were defined as clinically positive nodes, which included palpable ALNs on physical examination or suspicious ALNs detected by imaging, including CT. All patients underwent CNB or FNA of suspicious ALNs, which served as the pathological criterion for determining their status. The inclusion criteria were as follows: (1) pathology-confirmed invasive BC; (2) cN + status with subsequent CNB or FNA for suspicious ALNs; (3) complete high-resolution thin-section chest-enhanced CT examinations performed in the radiology department within one month prior to undergoing CNB or FNA; and (4) availability of complete clinicopathological data. The exclusion criteria were as follows: (1) confirmed diagnoses of bilateral primary or metastatic BC; (2) prior receipt of NAT before CNB or FNA; (3) incomplete CT scan sequences, poor imaging quality, completion of flat scans only, or CT performed at an outside institution; and (4) the presence of distant metastatic lesions or coexistence with other malignancies.

For the cN0 cohort, which included 450 patients, cN0 was defined as clinically ALN-negative, including negative physical examination and no suspicious ALNs detected by imaging, including CT. All patients underwent SLNB, with SLNB-positive patients potentially receiving ALND or radiation therapy. Surgical pathology was used to determine the ALN status. The inclusion criteria were as follows: (1) postoperative pathology confirming invasive BC; (2) cN0 status followed by

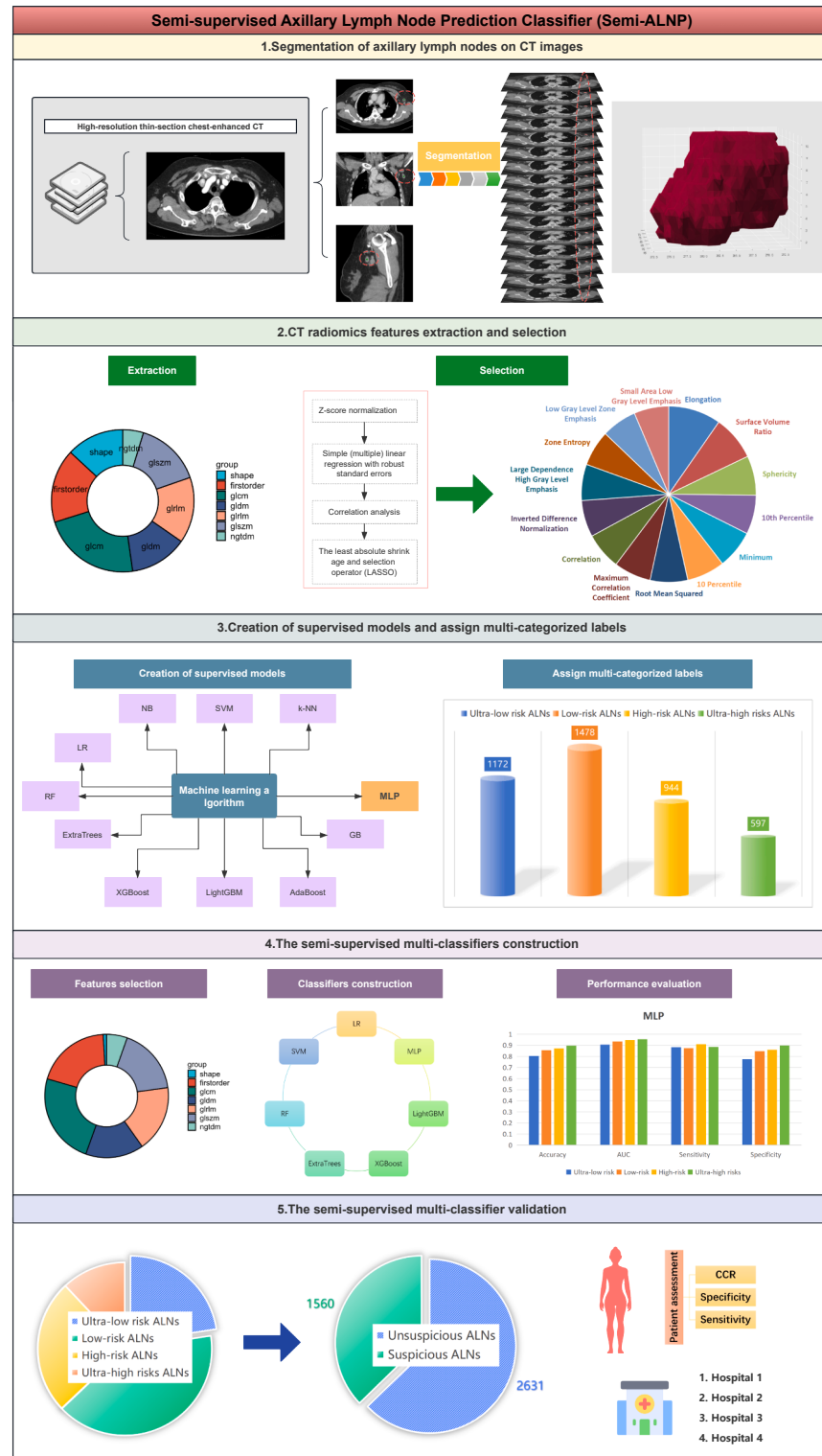


Fig. 1: Workflow diagram of this study.

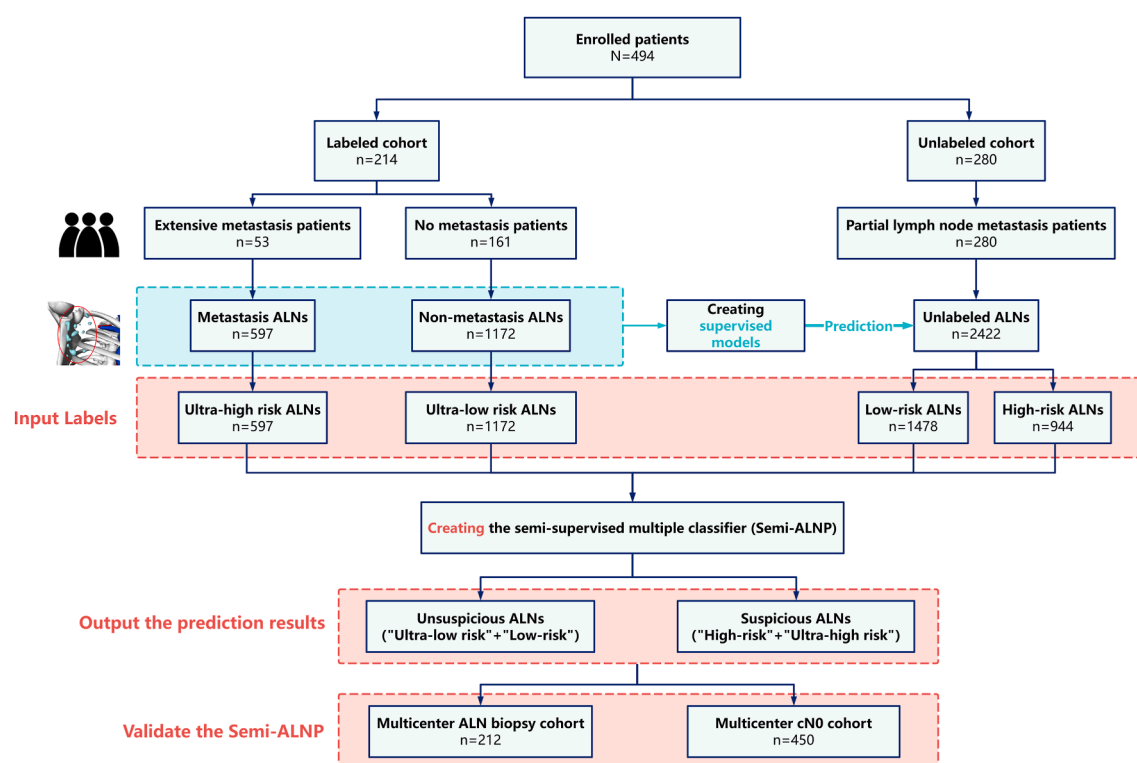


Fig. 2: Overall flow chart of this study.

SLNB; (3) complete chest-enhanced CT examinations conducted in the radiology department within one month prior to surgery; and (4) availability of complete clinicopathological data. The exclusion criteria were as follows: (1) confirmed diagnoses of bilateral primary or metastatic BC; (2) underwent NAT prior to surgery; (3) incomplete CT scan sequences, poor imaging quality, completion of flat scans only, or CT performed at an outside institution; and (4) the presence of distant metastatic lesions or coexistence with other malignancies.

## Ethics

This study was approved by the Institutional Ethics Review Boards of all participating hospitals. Specifically, ethical approval was granted by the Ethics Review Committee of Xiangya Second Hospital, Central South University (Approval No. LYF2023043), the Ethics Committee of the First People's Hospital of Xiangtan City (Approval No. 2023060601), the Ethics Committee of the First People's Hospital of Changde City (Approval No. 2025-228-01), and the Ethics Committee of the First People's Hospital of Chenzhou City (Approval No. 2025012). As this was a retrospective study, the requirement for informed consent was waived by all ethics committees.

## CT image acquisition

All enrolled patients received a high-resolution, thin-slice contrast-enhanced chest CT within one month prior to surgery, as determined by the clinical team. Scans were acquired using one of the following CT systems: (1) GE Revolution 256 or GE680 64; (2) Siemens Somatom Perspective, Definition Flash, or Force; and (3) United Imaging uCT780. Acquisition settings included 120 kV tube voltage, 200 mAs current, a 0.5-s rotation time, and detector collimation of  $64 \times 0.625$  mm. Images were reconstructed with a 1 mm slice thickness and 0 mm spacing, using a  $512 \times 512$  matrix and a field of view between 400 and 500 mm. Intravenous contrast (iopromide) was administered through the antecubital vein using a dual-head injector at 3.0 mL/s (1.2 mL/kg), followed by a 30 mL saline chaser.

## Segmentation of the ROIs of the ALNs

Axillary lymph node ROIs were delineated in ITK-SNAP v3.8.0 via a region-growing algorithm.<sup>37</sup> For each node, seed points were placed within the lymph node and automatically propagated through adjacent slices to capture its complete three-dimensional structure (Fig. 1). Two independent operators, blinded to all the clinical and pathological information, performed

the region-growing segmentation. All resulting ROIs were then reviewed and finalized by a senior radiologist to ensure anatomical accuracy and consistency.

### Definition of ALN labels

Histopathologic examination of ALND specimens provided the basis for ALN labeling. Nodes from patients with no metastatic involvement ( $n = 161$ ) were designated as ultra-low risk, while nodes from patients whose entire dissected ALNs were metastatic ( $n = 53$ ) were designated as ultra-high risk. ALNs from patients with partial metastatic involvement ( $n = 280$ ) were classified as unlabeled owing to the inability to achieve precise one-to-one mapping between CT-visible nodes and pathology. Isolated tumor cells (ITCs) were considered nonmetastatic.

Pseudolabels for the unlabeled cohort were generated by training a supervised classifier on the ultra-low and ultra-high risk groups. The optimal cutoff probability was determined via the Youden index, and each unlabeled ALN was subsequently assigned a “low risk” or “high risk” status on the basis of this threshold. To reduce label noise, a confidence-based filtering strategy was applied: nodes with predicted probabilities  $\geq 0.5$  or  $\leq 0.2$  were considered high-confidence and included for model training, whereas intermediate-confidence nodes were excluded from training but retained for internal validation. Although soft-labeling has theoretical advantages, we adopted this binary pseudolabeling scheme because of its greater interpretability and alignment with clinical decision-making. A patient-level validation—aggregating node-level predictions per patient—is presented in [Supplementary Table S8](#) to confirm the consistency and accuracy of this labeling strategy.

### Radiomic feature extraction and selection

Using PyRadiomics with Python 3.9.7 (<https://www.python.org/>), features were extracted from the ROI masks contained in the original CT images ([Supplementary Table S1](#)).<sup>38</sup> Before feature extraction, all the CT images were resampled to an isotropic voxel size of  $1 \text{ mm}^3$  ( $1 \times 1 \times 1 \text{ mm}$ ), and the gray-level histogram bin width was standardized to 25. To mitigate interpatient variability, including differences in contrast enhancement and CT acquisition parameters, all extracted radiomic features were standardized using Z-score normalization before model training.

In the supervised models, a total of 107 quantitative radiological features were extracted for each target ALN. These features can be subdivided into seven major categories ([Supplementary Fig. S1A](#)). The 107 features were first normalized, and least absolute shrinkage and selection operator (LASSO)-based logistic regression was then applied to reduce the number of features. Spearman correlation analysis was then performed, and correlation coefficients between the features were calculated to assess multicollinearity. If the correlation coefficient

of a pair of features was  $\geq 0.9$  or  $\leq -0.9$ , only the feature with the best diagnostic effect was retained. A total of 14 features were included in the training process of the supervised model ([Supplementary Fig. S1B–E](#)).

In the semisupervised multiclassifier, a total of 1561 quantitative radiological features were extracted for each target ALN, and these features were again subdivided into seven major categories ([Supplementary Fig. S2A](#)). Following normalization, LASSO-based logistic regression was used to reduce the number of features, resulting in 77 features being included in the training process of the semisupervised multiclassifier ([Supplementary Fig. S2B–E](#)).

### Prediction model construction and internal validation

The model construction and validation processes were conducted via Python 3.9.7 (<https://www.python.org/>). A total of 13 ML algorithms were applied, including logistic regression (LR), naive Bayes (NB), support vector machine (SVM), k-nearest neighbors (k-NN), random forest (RF), ExtraTrees, extreme gradient boosting (XGBoost), linear discriminant analysis (LDA), quadratic discriminant analysis (QDA), light gradient boosting machine (LightGBM), GradientBoosting, adaptive boosting (AdaBoost), and multilayer perceptron (MLP) methods ([Supplementary Table S2](#)).<sup>39</sup> Each classified model was trained to accurately fit the labeled data and predict the patients in the testing cohort. To choose the best hyperparameters for fitting the model, the grid search method and 5-fold cross-validation were performed. To avoid information leakage arising from the same patient contributing multiple ALNs, we employed a grouped cross-validation strategy via the GroupKFold method, in which ALNs were grouped by patient ID. This ensures that all ALNs from a single patient are included exclusively in either the training or validation set, maintaining the independence assumption between folds. To assess potential clustering effects due to multiple ALNs per patient, we quantified intraclass correlation coefficients (ICCs), which demonstrated negligible clustering ( $\text{ICC} = -0.002$ ), supporting the assumption of ALN-level independence ([Supplementary Table S23](#)). Four folds (80% of the ALNs, labeled and unlabeled) were used to train the model, and the remaining folds (20% of the ALNs, labeled and unlabeled) were used to select the best hyperparameters. The best-performing models from the primary cohort were subsequently tested on external validation cohorts.<sup>40</sup> Receiver operating characteristic (ROC) curves, the AUC, sensitivity, specificity, confusion matrices, and a decision curve analysis (DCA) were used to evaluate the efficacy of each model.

### Statistics

In this study, the baseline patient data were statistically analyzed via the R software (version 4.1.3) of R Studio

(R Project for Statistical Computing) and SPSS statistical software v27.0.<sup>41</sup> Continuous baseline variables were compared via linear regression, and the results are reported as the mean differences with 95% confidence intervals. Categorical variables were analyzed via logistic regression, with the results presented as odds ratios (or differences in proportions) and 95% confidence intervals. The confidence intervals for the AUC were calculated via Python's logistic regression function, which is based on the trapezoidal rule. For performance comparisons between models, DeLong's test was used to assess the statistical significance of pairwise AUC differences (Supplementary Table S18). For each model, discriminators such as the AUC, sensitivity, and specificity were calculated. The significance levels were two-sided, and comparisons between two or more groups were considered statistically significant if their P values were less than 0.05.

### Role of funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report. All investigators read, discussed, and approved the final version of this manuscript. All investigators had full access to the dataset and took responsibility for the authenticity and integrity of the dataset as well as the decision to submit for publication.

## Results

### Baseline characteristics of the enrolled patients

A total of 494 female BC patients were included in this study and divided into two cohorts: a labeled cohort and an unlabeled cohort (Fig. 2). The median age of the patients in the total cohort was 52 years, with 94.7% having invasive ductal carcinoma (IDC) and 68.6% having T2 tumors. The patients were molecularly classified on the basis of their hormone receptor, HER2, and Ki-67 statuses. The results indicated that 283 patients (57.3%) were classified as HR + HER2-, 123 patients (24.9%) were classified as HER2 positive, and 88 patients (17.8%) were classified as triple-negative breast cancer (TNBC). All patients underwent ALND surgery, and ALN staging was conducted on the basis of a pathological ALN examination. According to the presence or absence of ALN labels, the 494 patients were divided into cohorts with 214 labeled members and 280 unlabeled members (Table 1).

With postoperative pathology as the gold standard, the labeled cohort was divided into metastatic (n = 53) and nonmetastatic (n = 161) groups. The metastatic group included a total of 597 ALNs, whereas the non-metastatic group included 1172 ALNs. Based on group comparisons expressed as difference and 95% confidence intervals, several clinicopathological factors demonstrated statistically significant differences between the two groups. Specifically, the T-category, ER

|                            | Overall<br>(N = 494) | Labeled cohort<br>(N = 214) | Unlabeled cohort<br>(N = 280) |
|----------------------------|----------------------|-----------------------------|-------------------------------|
| <b>Age</b>                 |                      |                             |                               |
| Mean (SD)                  | 52.4 (10.7)          | 52.9 (10.5)                 | 52.0 (10.8)                   |
| Median [Min, Max]          | 52 [21, 81]          | 53 [21, 81]                 | 52 [21, 81]                   |
| <b>Pathological types</b>  |                      |                             |                               |
| IDC                        | 468 (94.7%)          | 201 (93.9%)                 | 268 (95.7%)                   |
| ILC                        | 20 (4.0%)            | 11 (5.1%)                   | 9 (3.2%)                      |
| IMC                        | 3 (0.6%)             | 1 (0.5%)                    | 1 (0.4%)                      |
| IPC                        | 3 (0.6%)             | 1 (0.5%)                    | 2 (0.8%)                      |
| <b>Histological grades</b> |                      |                             |                               |
| I                          | 6 (1.2%)             | 4 (1.9%)                    | 2 (0.7%)                      |
| II                         | 390 (78.9%)          | 168 (78.5%)                 | 222 (79.3%)                   |
| III                        | 98 (19.8%)           | 42 (19.6%)                  | 56 (20.0%)                    |
| <b>T category</b>          |                      |                             |                               |
| T1                         | 126 (25.5%)          | 66 (30.8%)                  | 60 (21.4%)                    |
| T2                         | 339 (68.6%)          | 136 (63.6%)                 | 203 (72.5%)                   |
| T3                         | 29 (5.9%)            | 12 (5.6%)                   | 17 (6.1%)                     |
| <b>ER</b>                  |                      |                             |                               |
| Negative                   | 157 (31.8%)          | 76 (35.5%)                  | 81 (28.9%)                    |
| Positive                   | 337 (68.2%)          | 138 (64.5%)                 | 199 (71.1%)                   |
| <b>PR</b>                  |                      |                             |                               |
| Negative                   | 194 (39.3%)          | 94 (43.9%)                  | 100 (35.7%)                   |
| Positive                   | 300 (60.7%)          | 120 (56.1%)                 | 180 (64.3%)                   |
| <b>HER-2</b>               |                      |                             |                               |
| Negative                   | 371 (75.1%)          | 156 (72.9%)                 | 215 (76.8%)                   |
| Positive                   | 123 (24.9%)          | 58 (27.1%)                  | 65 (23.2%)                    |
| <b>Ki-67</b>               |                      |                             |                               |
| <14                        | 79 (16.0%)           | 41 (19.2%)                  | 38 (13.6%)                    |
| ≥14                        | 415 (84.0%)          | 173 (80.8%)                 | 242 (86.4%)                   |
| <b>Subtypes</b>            |                      |                             |                               |
| HER2+                      | 123 (24.9%)          | 58 (27.1%)                  | 65 (23.2%)                    |
| HR + HER2-                 | 283 (57.3%)          | 116 (54.2%)                 | 167 (59.6%)                   |
| TNBC                       | 88 (17.8%)           | 40 (18.7%)                  | 48 (17.2%)                    |

IDC, invasive ductal carcinoma; ILC, invasive lobular carcinoma; IMC, invasive mucinous carcinoma; IPC, invasive papillary carcinoma; TNBC, triple-negative breast cancer; ER, estrogen receptor; PR, progesterone receptor; HER-2, human epidermal growth factor receptor 2.

**Table 1: Baseline characteristics of each cohort.**

status, PR status, and molecular subtypes showed notable differences, with larger tumors, hormone receptor negativity, and TNBC being more frequently associated with ALN metastasis. In contrast, age, pathological type, histological grade, HER2 status, and Ki-67 status did not show significant differences between the groups (Table 2).

In the unlabeled cohort consisting of 280 patients, all patients had ALNM, with the highest percentage represented by N1 patients (61.8%). A total of 2422 ALNs were segmented on the CT images of these patients. Accurate labels could not be obtained because these ALNs could not be matched to the pathological examination results (Table 1).

### Creation of supervised models

To further address the labeling issue posed by the 2422 unlabeled ALNs, we first built a supervised model based

|                            | Metastasis<br>(N = 53) | No metastasis<br>(N = 161) | Overall<br>(N = 214) | Difference (95% CI)     |
|----------------------------|------------------------|----------------------------|----------------------|-------------------------|
| <b>Age</b>                 |                        |                            |                      |                         |
| Mean (SD)                  | 53.4 (10.6)            | 52.8 (10.5)                | 52.9 (10.5)          | 0.6 (−2.7, 3.9)         |
| Median [Min, Max]          | 53 [33, 81]            | 53 [21, 81]                | 53 [21, 81]          |                         |
| <b>Pathological types</b>  |                        |                            |                      |                         |
| IDC                        | 50 (94.3%)             | 151 (93.8%)                | 201 (93.9%)          | 0.5% (−3.6%, 4.6%)      |
| ILC                        | 3 (5.7%)               | 8 (5.0%)                   | 11 (5.1%)            | 0.7% (−3.2%, 4.6%)      |
| IMC                        | 0 (0%)                 | 1.00 (0.6%)                | 1 (0.5%)             | −0.6% (−1.7%, 0.5%)     |
| IPC                        | 0 (0%)                 | 1.00 (0.6%)                | 1 (0.5%)             | −0.6% (−1.7%, 0.5%)     |
| <b>Histological grades</b> |                        |                            |                      |                         |
| I                          | 0 (0%)                 | 4 (2.5%)                   | 4 (1.9%)             | −2.5% (−5.6%, 0.6%)     |
| II                         | 41 (77.4%)             | 127 (78.9%)                | 168 (78.5%)          | −1.5% (−11.2%, 8.2%)    |
| III                        | 12 (22.6%)             | 30 (18.6%)                 | 42 (19.6%)           | 4.0% (−4.7%, 12.7%)     |
| <b>T category</b>          |                        |                            |                      |                         |
| T1                         | 7 (13.2%)              | 59 (36.6%)                 | 66 (30.8%)           | −23.4% (−34.5%, −12.3%) |
| T2                         | 38 (71.7%)             | 98 (60.9%)                 | 136 (63.6%)          | 10.8% (1.2%, 20.4%)     |
| T3                         | 8 (15.1%)              | 4 (2.5%)                   | 12 (5.6%)            | 12.6% (6.3%, 18.9%)     |
| <b>ER</b>                  |                        |                            |                      |                         |
| Negative                   | 15 (28.3%)             | 61 (37.9%)                 | 76 (35.5%)           | −9.6% (−18.0%, −1.2%)   |
| Positive                   | 38 (71.7%)             | 100 (62.1%)                | 138 (64.5%)          | 9.6% (1.2%, 18.0%)      |
| <b>PR</b>                  |                        |                            |                      |                         |
| Negative                   | 17 (32.1%)             | 77 (47.8%)                 | 94 (43.9%)           | −15.7% (−27.2%, −4.2%)  |
| Positive                   | 36 (67.9%)             | 84 (52.2%)                 | 120 (56.1%)          | 15.7% (4.2%, 27.2%)     |
| <b>HER-2</b>               |                        |                            |                      |                         |
| Negative                   | 40 (75.5%)             | 116 (72.0%)                | 156 (72.9%)          | 3.5% (−7.2%, 14.2%)     |
| Positive                   | 13 (24.5%)             | 45 (28.0%)                 | 58 (27.1%)           | −3.5% (−14.2%, 7.2%)    |
| <b>Ki-67</b>               |                        |                            |                      |                         |
| <14                        | 11 (20.8%)             | 30 (18.6%)                 | 41 (19.2%)           | 2.2% (−7.8%, 12.2%)     |
| ≥14                        | 42 (79.2%)             | 131 (81.4%)                | 173 (80.8%)          | −2.2% (−12.2%, 7.8%)    |
| <b>Subtypes</b>            |                        |                            |                      |                         |
| HER2+                      | 13 (24.5%)             | 45 (28.0%)                 | 58 (27.1%)           | −3.5% (−14.2%, 7.2%)    |
| HR + HER2−                 | 33 (62.3%)             | 83 (51.5%)                 | 116 (54.2%)          | 10.8% (0.5%, 21.1%)     |
| TNBC                       | 7 (13.2%)              | 33 (20.5%)                 | 40 (18.7%)           | −7.3% (−16.8%, 2.2%)    |

Difference (95% CI) calculated as: Metastatic-Non-metastatic; IDC, invasive ductal carcinoma; ILC, invasive lobular carcinoma; IMC, invasive mucinous carcinoma; IPC, invasive papillary carcinoma; TNBC, triple-negative breast cancer; ER, estrogen receptor; PR, progesterone receptor; HER-2, human epidermal growth factor receptor 2.

**Table 2: Baseline characteristics of the BC patients in the nonmetastatic and metastatic ALN groups of the labeled cohort.**

on the 1769 labeled ALNs derived from the labeled cohort (214 patients). 11 supervised ML classification algorithms were applied to construct the models (Table 3).

Model selection was guided by a comprehensive comparison of the AUC and sensitivity across all the candidate models (Fig. 3A and B, Supplementary Table S18). The MLP model ranked first in both AUC (0.959) and sensitivity (0.899), demonstrating a superior balance between diagnostic accuracy and the ability to identify positive ALNs (Supplementary Fig. S3 and Table S15). The high performance of the MLP is likely attributable to its deep learning architecture, which allows for better fitting of nonlinear patterns and subtle radiomic variations inherent in the data (Supplementary Fig. S4). A further evaluation of the MLP model revealed significant numbers of true positives and true negatives in the confusion matrix (Fig. 3C and D). A decision curve analysis (DCA) demonstrated that the net benefit curve of the model surpassed both the “treat-all” curve (representing the scenario where all patients are assumed to have ALNM and are treated accordingly) and the “treat-none” curve (representing the scenario where no patients are assumed to have ALNM and none are treated). This further validated the model’s utility in clinical decision-making scenarios (Fig. 3E and F). In addition, the radiomics nomogram scores obtained for each ALN indicated that the model had strong discriminatory power (Fig. 3G and H). Considering these findings, the MLP model was selected as the final supervised model. It was subsequently employed to predict the 2422 unlabeled ALNs, generating a metastatic prediction probability for each ALN. These predicted values were then used to assign pseudolabels as part of the semi-supervised learning process.

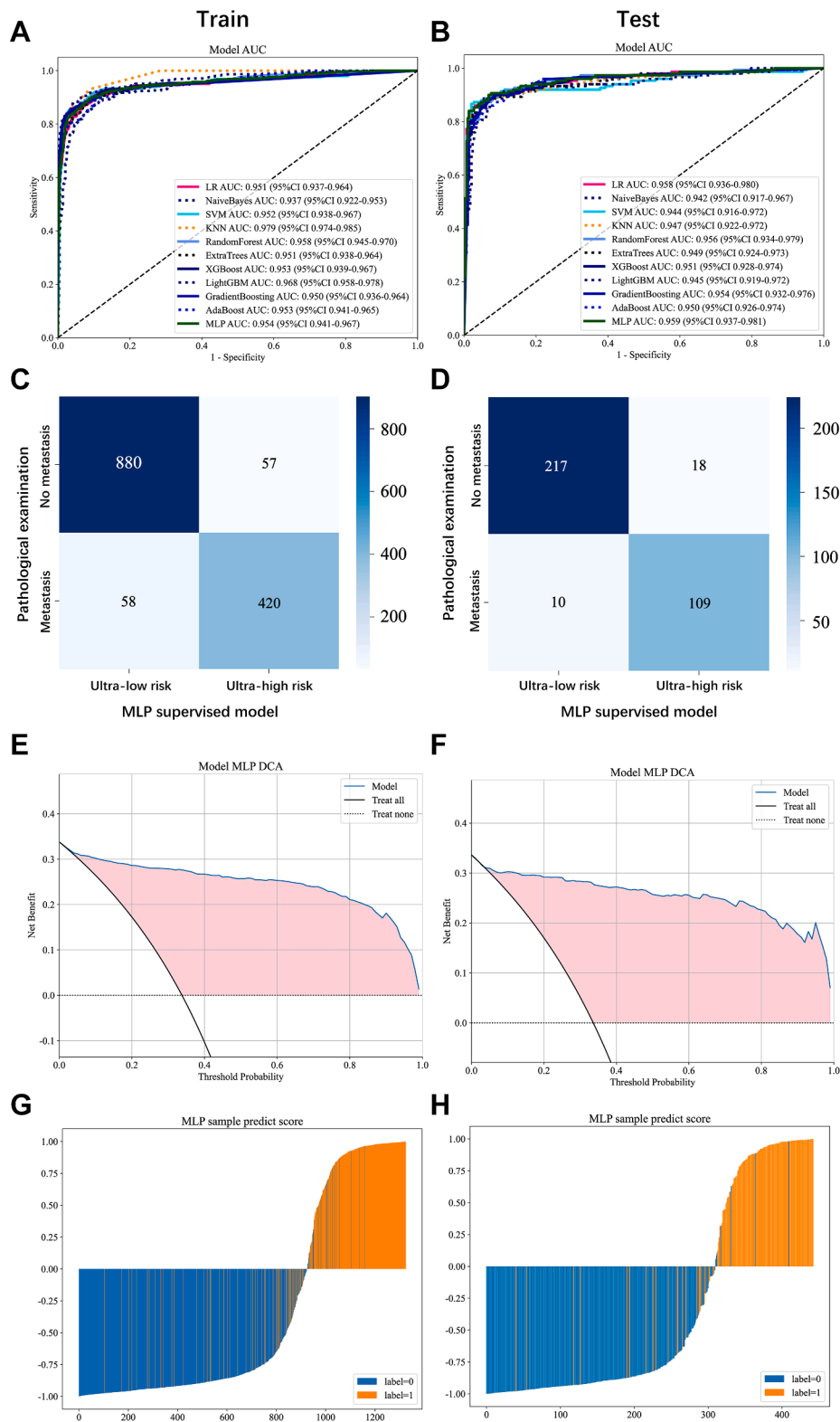
### Assigning multicategorized labels

After obtaining the predicted probabilities for the unlabeled ALNs, we sequentially assigned labels to the

| Models     | AUC                 | Sensitivity         | Specificity         | Accuracy            | Threshold |
|------------|---------------------|---------------------|---------------------|---------------------|-----------|
| LR         | 0.958 (0.937–0.980) | 0.853 (0.792–0.906) | 0.976 (0.957–0.993) | 0.934 (0.910–0.957) | 0.506     |
| NB         | 0.942 (0.917–0.967) | 0.886 (0.834–0.933) | 0.922 (0.891–0.954) | 0.910 (0.883–0.937) | 0.175     |
| SVM        | 0.944 (0.916–0.972) | 0.867 (0.809–0.921) | 0.980 (0.962–0.993) | 0.942 (0.919–0.964) | 0.379     |
| k-NN       | 0.947 (0.922–0.972) | 0.859 (0.804–0.913) | 0.969 (0.949–0.989) | 0.932 (0.910–0.955) | 0.600     |
| RF         | 0.956 (0.934–0.979) | 0.865 (0.810–0.915) | 0.956 (0.930–0.977) | 0.925 (0.901–0.948) | 0.392     |
| ExtraTrees | 0.949 (0.924–0.973) | 0.886 (0.830–0.933) | 0.932 (0.902–0.959) | 0.916 (0.892–0.941) | 0.343     |
| XGBoost    | 0.951 (0.928–0.974) | 0.886 (0.831–0.936) | 0.935 (0.902–0.962) | 0.918 (0.889–0.944) | 0.348     |
| LightGBM   | 0.945 (0.919–0.972) | 0.886 (0.834–0.934) | 0.945 (0.918–0.970) | 0.925 (0.898–0.948) | 0.350     |
| GB         | 0.954 (0.932–0.976) | 0.861 (0.805–0.912) | 0.952 (0.926–0.976) | 0.921 (0.894–0.946) | 0.364     |
| AdaBoost   | 0.950 (0.926–0.974) | 0.846 (0.792–0.897) | 0.941 (0.916–0.966) | 0.909 (0.883–0.932) | 0.494     |
| MLP        | 0.959 (0.937–0.981) | 0.899 (0.847–0.939) | 0.932 (0.903–0.958) | 0.921 (0.896–0.944) | 0.319     |

Note: 95% confidence intervals are provided in parentheses.

**Table 3: Comparison of the AUC values and predictive efficacy metrics yielded by the supervised models.**



**Fig. 3:** Efficacy assessment of the supervised models. A and B: Summary of the ROC curves and AUC values produced by the supervised models. C and D: Confusion matrices yielded by the MLP model for the training and test sets. E and F: DCA of the MLP model on the training and test sets. G and H: Radiomics nomogram score produced for each ALN. The blue bars represent the scores of the nonmetastatic ALNs, whereas the orange bars represent the scores of the metastatic ALNs.

4191 enrolled ALNs. In the labeled cohort, 1172 ALNs derived from 161 patients were pathologically confirmed as nonmetastatic, these ALNs were therefore classified as “ultra-low risk”. Among the 597 ALNs derived from 53 patients, all showed metastasis during the pathological examination and were therefore labeled “ultra-high risk.” In the unlabeled cohort, pseudolabels were assigned on the basis of predictions from the MLP model, using the optimal cutoff determined by the Youden index. Specifically, 1478 ALNs with predicted metastatic probabilities below the threshold were labeled “low risk,” whereas 944 ALNs with probabilities above the threshold were labeled “high risk.” (Fig. 2).

### Performance of semisupervised multiclassifiers

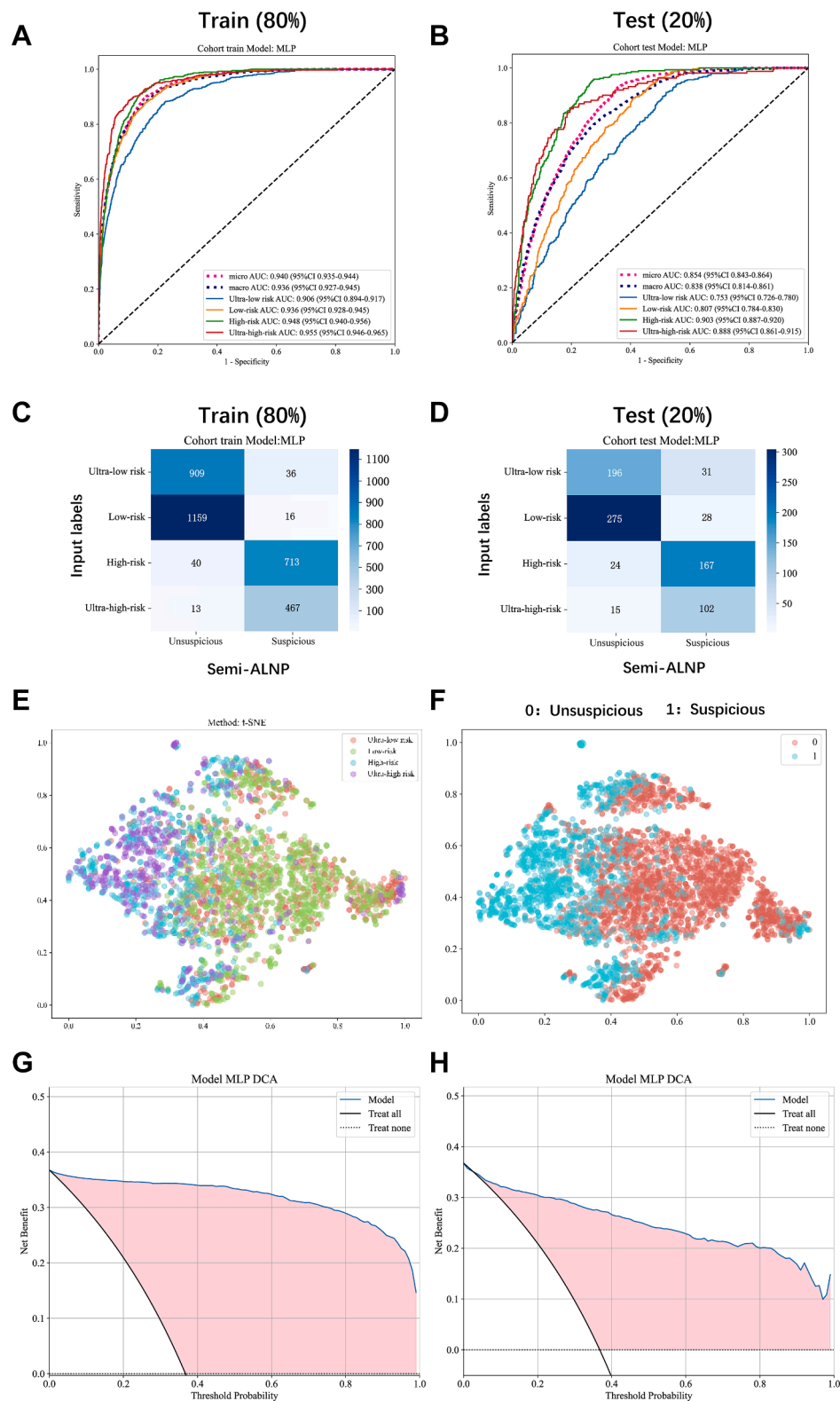
By segmenting the CT images of 4191 ALNs, a total of 1561 radiomic features were extracted. After performing feature selection, 77 radiomic features were ultimately chosen to construct semisupervised multiclassifiers (Supplementary Fig. S2). Seven ML algorithms were employed for the development of these multiclassifiers. Table 4 lists the diagnostic capabilities of these seven different models on the training set. The

AUC (macro/micro) values obtained for the training cohort were as follows for the seven models: 0.800/0.794 for LR, 0.873/0.861 for the SVM, 0.814/0.790 for the random forest, 0.770/0.759 for ExtraTrees, 0.856/0.845 for XGBoost, 0.824/0.839 for the LightGBM, and 0.940/0.936 for the MLP (Fig. 4A and B, Supplementary Fig. S5). A comparison of the diagnostic efficacy levels of these seven multiclassifiers revealed that the MLP model outperformed the other models. The AUCs for the ultra-low risk, low-risk, high-risk, and ultra-high-risk classes were 0.906, 0.936, 0.948, and 0.955, respectively. The corresponding sensitivities were 0.883, 0.875, 0.910, and 0.886, whereas the specificities were 0.775, 0.847, 0.860, and 0.899, respectively (Table 4). Confusion matrices were applied to the training and testing cohorts, and seven models were ranked by AUC and sensitivity, with the MLP model achieving the highest predictive accuracy (Fig. 4C and D, Supplementary Table S15). Based on these findings, the MLP model was named Semi-ALNP for its application in the subsequent analysis of ALNs.

Next, the radiomic features were downsampled to two dimensions to observe the distribution of differently

| Model          | Label           | Accuracy            | AUC                 | Sensitivity         | Specificity          |
|----------------|-----------------|---------------------|---------------------|---------------------|----------------------|
| LR             | Ultra-low risk  | 0.623 (0.616–0.676) | 0.729 (0.710–0.747) | 0.804 (0.777–0.836) | 0.660 (0.649–0.688)  |
|                | Low-risk        | 0.692 (0.683–0.724) | 0.776 (0.760–0.793) | 0.913 (0.889–0.932) | 0.576 (0.557–0.593)  |
|                | High-risk       | 0.771 (0.744–0.799) | 0.866 (0.853–0.879) | 0.890 (0.856–0.911) | 0.735 (0.714–0.767)  |
|                | Ultra-high risk | 0.734 (0.699–0.764) | 0.803 (0.782–0.825) | 0.705 (0.670–0.753) | 0.738 (0.708–0.749)  |
| SVM            | Ultra-low risk  | 0.664 (0.648–0.84)  | 0.788 (0.771–0.804) | 0.824 (0.803–0.848) | 0.601 (0.575–0.646)  |
|                | Low-risk        | 0.731 (0.685–0.782) | 0.831 (0.816–0.845) | 0.898 (0.860–0.923) | 0.644 (0.606–0.686)  |
|                | High-risk       | 0.801 (0.788–0.836) | 0.907 (0.896–0.918) | 0.904 (0.880–0.928) | 0.770 (0.751–0.793)  |
|                | Ultra-high risk | 0.851 (0.801–0.906) | 0.919 (0.906–0.933) | 0.860 (0.849–0.878) | 0.850 (0.829–0.885)  |
| RF             | Ultra-low risk  | 0.579 (0.558–0.594) | 0.698 (0.680–0.716) | 0.873 (0.866–0.886) | 0.461 (0.438–0.493)  |
|                | Low-risk        | 0.684 (0.682–0.696) | 0.774 (0.759–0.790) | 0.900 (0.888–0.936) | 0.571 (0.563–0.578)  |
|                | High-risk       | 0.757 (0.743–0.769) | 0.842 (0.828–0.857) | 0.896 (0.863–0.926) | 0.716 (0.693–0.740)  |
|                | Ultra-high risk | 0.786 (0.775–0.796) | 0.846 (0.826–0.865) | 0.798 (0.772–0.813) | 0.784 (0.774–0.790)  |
| ExtraTrees     | Ultra-low risk  | 0.598 (0.537–0.655) | 0.693 (0.673–0.713) | 0.783 (0.752–0.818) | 0.525 (0.480–0.579)  |
|                | Low-risk        | 0.655 (0.630–0.683) | 0.740 (0.722–0.758) | 0.822 (0.810–0.841) | 0.568 (0.551–0.583)  |
|                | High-risk       | 0.719 (0.702–0.735) | 0.809 (0.794–0.825) | 0.843 (0.813–0.868) | 0.682 (0.671–0.709)  |
|                | Ultra-high risk | 0.722 (0.702–0.747) | 0.793 (0.772–0.815) | 0.751 (0.723–0.787) | 0.717 (0.701–0.731)  |
| XGBoost        | Ultra-low risk  | 0.651 (0.637–0.680) | 0.768 (0.751–0.785) | 0.828 (0.794–0.849) | 0.580 (0.550–0.602)  |
|                | Low-risk        | 0.709 (0.699–0.724) | 0.822 (0.808–0.837) | 0.919 (0.871–0.969) | 0.588 (0.543–0.618)  |
|                | High-risk       | 0.771 (0.748–0.801) | 0.893 (0.881–0.905) | 0.934 (0.901–0.950) | 0.723 (0.710–0.745)  |
|                | Ultra-high risk | 0.807 (0.782–0.843) | 0.897 (0.882–0.913) | 0.872 (0.814–0.939) | 0.797 (0.764–0.824)  |
| LightGBM       | Ultra-low risk  | 0.689 (0.668–0.711) | 0.762 (0.745–0.780) | 0.727 (0.696–0.749) | 0.688 (0.616–0.740)  |
|                | Low-risk        | 0.698 (0.638–0.747) | 0.819 (0.804–0.833) | 0.944 (0.928–0.964) | 0.569 (0.555–0.577)  |
|                | High-risk       | 0.791 (0.751–0.843) | 0.884 (0.871–0.896) | 0.884 (0.855–0.923) | 0.764 (0.745–0.779)  |
|                | Ultra-high risk | 0.765 (0.746–0.809) | 0.890 (0.874–0.906) | 0.900 (0.887–0.923) | 0.743 (0.701–0.788)  |
| MLP(Semi-ALNP) | Ultra-low risk  | 0.805 (0.739–0.882) | 0.906 (0.894–0.917) | 0.883 (0.867–0.913) | 0.775 (0.745–0.803)  |
|                | Low-risk        | 0.856 (0.839–0.880) | 0.936 (0.928–0.945) | 0.875 (0.838–0.925) | 0.847 (0.8153–0.869) |
|                | High-risk       | 0.871 (0.868–0.880) | 0.948 (0.940–0.956) | 0.910 (0.860–0.971) | 0.860 (0.848–0.878)  |
|                | Ultra-high risk | 0.897 (0.885–0.914) | 0.955 (0.946–0.965) | 0.886 (0.860–0.915) | 0.899 (0.883–0.919)  |

Table 4: Comparison of the AUC values and predictive efficacy metrics of semisupervised multiclassifiers.



**Fig. 4:** Efficacy assessment of Semi-ALNP. A and B: Summary of the ROC curves and AUC values obtained for the semisupervised MLP multiclassifier in the training set (80%) and test set (20%). C and D: Confusion matrix yielded by the Semi-ALNP on the training set (80%) and test set (20%). E and F: Distributions predicted by the Semi-ALNP for differently labeled samples in 2D space. G and H: DCA of the Semi-ALNP on the training and test sets.

|                            | The Second Xiangya Hospital (N = 144) | The First People's Hospital of Chenzhou city (N = 31) | The First People's Hospital of Changde city (N = 30) | The First People's Hospital of Xiangtan city (N = 7) | Overall (N = 212) |
|----------------------------|---------------------------------------|-------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-------------------|
| <b>Age</b>                 |                                       |                                                       |                                                      |                                                      |                   |
| Mean (SD)                  | 52.2 (10.30)                          | 53.2 (9.14)                                           | 55.6 (11.6)                                          | 59.4 (16.0)                                          | 53.1 (10.6)       |
| Median [Min, Max]          | 52 [27, 81]                           | 53 [36, 67]                                           | 54 [31, 87]                                          | 53 [36, 82]                                          | 52 [27, 87]       |
| <b>Pathological types</b>  |                                       |                                                       |                                                      |                                                      |                   |
| IDC                        | 141 (97.9%)                           | 29 (93.5%)                                            | 27 (90.0%)                                           | 5 (71.4%)                                            | 202 (95.3%)       |
| ILC                        | 3 (2.1%)                              | 1 (3.2%)                                              | 2 (6.7%)                                             | 2 (28.6%)                                            | 8 (3.8%)          |
| IPC                        | 0 (0%)                                | 1 (3.2%)                                              | 1 (3.3%)                                             | 0 (0%)                                               | 2 (0.9%)          |
| <b>Histological grades</b> |                                       |                                                       |                                                      |                                                      |                   |
| I                          | 1 (0.7%)                              | 2 (6.5%)                                              | 3 (10.0%)                                            | 0 (0%)                                               | 6 (2.8%)          |
| II                         | 120 (83.3%)                           | 19 (61.3%)                                            | 10 (33.3%)                                           | 7 (100%)                                             | 156 (73.6%)       |
| III                        | 23 (16.0%)                            | 10 (32.3%)                                            | 17 (56.7%)                                           | 0 (0%)                                               | 50 (23.6%)        |
| <b>T category</b>          |                                       |                                                       |                                                      |                                                      |                   |
| T1                         | 18 (12.5%)                            | 6 (19.4%)                                             | 11 (36.7%)                                           | 2 (28.6%)                                            | 37 (17.5%)        |
| T2                         | 92 (63.9%)                            | 23 (74.2%)                                            | 18 (60.0%)                                           | 1 (14.3%)                                            | 134 (63.2%)       |
| T3                         | 22 (15.3%)                            | 2 (6.5%)                                              | 1 (3.3%)                                             | 3 (42.9%)                                            | 28 (13.2%)        |
| T4                         | 12 (8.3%)                             | 0 (0%)                                                | 0 (0%)                                               | 1 (14.3%)                                            | 13 (6.1%)         |
| <b>ALN biopsies</b>        |                                       |                                                       |                                                      |                                                      |                   |
| Negative                   | 24 (16.7%)                            | 4 (12.9%)                                             | 15 (50.0%)                                           | 4 (57.1%)                                            | 47 (22.2%)        |
| Positive                   | 120 (83.3%)                           | 27 (87.1%)                                            | 15 (50.0%)                                           | 3 (42.9%)                                            | 165 (77.8%)       |
| <b>ER</b>                  |                                       |                                                       |                                                      |                                                      |                   |
| Negative                   | 59 (41.0%)                            | 12 (38.7%)                                            | 7 (23.3%)                                            | 2 (28.6%)                                            | 80 (37.7%)        |
| Positive                   | 85 (59.0%)                            | 19 (61.3%)                                            | 23 (76.7%)                                           | 5 (71.4%)                                            | 132 (62.3%)       |
| <b>PR</b>                  |                                       |                                                       |                                                      |                                                      |                   |
| Negative                   | 68 (47.2%)                            | 14 (45.2%)                                            | 8 (26.7%)                                            | 3 (42.9%)                                            | 93 (43.9%)        |
| Positive                   | 76 (52.8%)                            | 17 (54.8%)                                            | 22 (73.3%)                                           | 4 (57.1%)                                            | 119 (56.1%)       |
| <b>HER-2</b>               |                                       |                                                       |                                                      |                                                      |                   |
| Negative                   | 81 (56.3%)                            | 24 (77.4%)                                            | 19 (63.3%)                                           | 6 (85.7%)                                            | 130 (61.3%)       |
| Positive                   | 63 (43.8%)                            | 7 (22.6%)                                             | 11 (36.7%)                                           | 1 (14.3%)                                            | 82 (38.7%)        |
| <b>Ki-67</b>               |                                       |                                                       |                                                      |                                                      |                   |
| <14                        | 7 (4.9%)                              | 9 (29.0%)                                             | 1 (3.3%)                                             | 0 (0%)                                               | 17 (8.0%)         |
| ≥14                        | 137 (95.1%)                           | 22 (71.0%)                                            | 29 (96.7%)                                           | 7 (100%)                                             | 195 (92.0%)       |
| <b>Subtypes</b>            |                                       |                                                       |                                                      |                                                      |                   |
| HER2+                      | 63 (43.7%)                            | 7 (22.6%)                                             | 11 (36.7%)                                           | 1 (14.3%)                                            | 82 (38.7%)        |
| HR + HER2-                 | 59 (41%)                              | 17 (54.9%)                                            | 17 (56.6%)                                           | 4 (57.1%)                                            | 97 (45.7%)        |
| TNBC                       | 22.0 (15.3%)                          | 7 (22.6%)                                             | 2 (6.7%)                                             | 2 (28.6%)                                            | 33 (15.6%)        |

IDC, invasive ductal carcinoma; ILC, invasive lobular carcinoma; IPC, invasive papillary carcinoma; TNBC, triple-negative breast cancer; ER, estrogen receptor; PR, progesterone receptor; HER-2, human epidermal growth factor receptor 2.

**Table 5: Baseline characteristics of the multicenter ALN biopsy cohort.**

|                  | Pathological examination |                |       | CCR              | Sensitivity      | Specificity    | FPR           | FNR           |
|------------------|--------------------------|----------------|-------|------------------|------------------|----------------|---------------|---------------|
|                  | Metastasis               | Non-metastasis | Total |                  |                  |                |               |               |
| Clinical imaging |                          |                |       |                  |                  |                |               |               |
| Suspicious       | 165                      | 47             | 212   | 77.83% (165/212) | 100% (165/165)   | 0% (0/47)      | 100% (47/47)  | 0% (0/165)    |
| Unsuspicious     | 0                        | 0              | 0     |                  |                  |                |               |               |
| Total            | 165                      | 47             | 212   |                  |                  |                |               |               |
| Semi-ALNP        |                          |                |       |                  |                  |                |               |               |
| Suspicious       | 163                      | 7              | 170   | 95.75% (203/212) | 98.79% (163/165) | 85.11% (40/47) | 14.89% (7/47) | 1.21% (2/165) |
| Unsuspicious     | 2                        | 40             | 42    |                  |                  |                |               |               |
| Total            | 165                      | 47             | 212   |                  |                  |                |               |               |

Semi-ALNP, semisupervised MLP multiclassifier; CCR, correct classification rate; FPR, false-positive rate; FNR, false-negative rate.

**Table 6: Comparison of the diagnostic efficacy of clinical imaging evaluation (ultrasound/mammography/MRI) and Semi-ALNP in the axillary biopsy cohort.**

**Table 6: Comparison of the diagnostic efficacy of clinical imaging evaluation (ultrasound/mammography/MRI) and Semi-ALNP in the axillary biopsy cohort.**

labeled samples. Notably, the ‘ultra-low risk’ label predicted by the Semi-ALNP closely aligned with the ‘low-risk’ label, whereas the ‘high-risk’ label was near the ‘ultra-high risk’ label in the same space (Fig. 4E and F). Based on these observations, ‘ultra-low risk’ and ‘low-risk’ ALNs were classified as ‘unsuspicious,’ while ‘high-risk’ and ‘ultra-high risk’ ALNs were categorized as ‘suspicious’ (Fig. 2). Decision curve analysis confirmed that this binary stratification offers practical net benefit across clinically relevant threshold probabilities (Fig. 4G and H).

To further validate accuracy at the patient level, node-level predictions were aggregated in the training and internal validation cohorts ( $n = 494$ ), yielding a correct classification rate (CCR) of 89.47% (442/494), sensitivity of 90.39% (301/333), and specificity of 87.58% (141/161) (Supplementary Table S8). Finally, compared with standard clinical assessments—ultrasound, mammography, and MRI—Semi-ALNP demonstrated superior CCR, sensitivity, and specificity (Supplementary Tables S3–S6), underscoring its potential to enhance routine decision-making.

### Multicenter validation of the Semi-ALNP in ALN biopsy and cN0 cohorts

To evaluate the diagnostic efficacy of the Semi-ALNP, we conducted a multicenter validation study across two cohorts: an ALN biopsy cohort (212 patients) and a cN0 cohort (450 patients).

Among the 212 patients who underwent percutaneous ALN biopsies, 77.8% (165/212) were positive ( $pN_{(b)}$ ) and 22.2% (47/212) were negative ( $pN_{(b)}$ ) based on needle biopsy results (CNB or FNA). The Semi-ALNP correctly predicted 163 out of 165  $cN_{+}/pN_{(b)}$  ALNs, achieving a sensitivity of 98.79%. It also correctly identified 40 out of 47  $cN_{+}/pN_{(b)}$  ALNs as unsuspicious, resulting in a specificity of 85.11% and a correct classification rate (CCR) of 95.75%, outperforming clinical imaging evaluation (CCR 77.8%) (Tables 5 and 6).

Among the 450 cN0 patients, all underwent SLNB, 76.0% (342/450) had no metastasis ( $pN0_{(sn)}$ ), 24.0% (108/450) had metastasis ( $pN_{+}(sn)$ ), and  $pN_{+}(sn)$  patients may opt for ALND (99/108) or radiation therapy (9/108). The Semi-ALNP accurately identified 308 out of 342  $cN0/pN0_{(sn)}$  patients as unsuspicious, achieving a specificity of 90.06%. Among the 108  $cN0/pN_{+}(sn)$  patients, it detected suspicious ALNs in 99 patients, resulting in a sensitivity of 91.67%. The overall CCR for the cN0 cohort was 90.44%, again exceeding clinical imaging (CCR 76.0%) (Tables 7 and 8).

To further assess robustness, subgroup analyses were performed. In the biopsy cohort, highest CCRs were observed in the HR+/HER2– (96.9%) and T2 (97.8%) subgroups, whereas the TNBC and T3–T4 subgroups achieved 100% sensitivity; no significant age-related differences were noted (<52 vs. ≥52 years) (Supplementary Tables S9–S11). In the cN0 cohort, the TNBC (93.4%), T1 (96.3%), and age < 52 (91.9%)

|                            | The Second Xiangya Hospital (N = 322) | The First People's Hospital of Chenzhou city (N = 115) | The First People's Hospital of Xiangtan city (N = 113) | Overall (N = 450) |
|----------------------------|---------------------------------------|--------------------------------------------------------|--------------------------------------------------------|-------------------|
| <b>Age</b>                 |                                       |                                                        |                                                        |                   |
| Mean (SD)                  | 51.8 (10.1)                           | 52.8 (10.4)                                            | 55.7 (9.27)                                            | 52.2 (10.2)       |
| Median [Min, Max]          | 52 [26, 81]                           | 52.0 [34, 79]                                          | 55 [36, 70]                                            | 52 [26, 81]       |
| <b>Pathological types</b>  |                                       |                                                        |                                                        |                   |
| IDC                        | 300 (93.2%)                           | 107 (93.0%)                                            | 12 (92.3%)                                             | 419 (93.1%)       |
| ILC                        | 12 (3.7%)                             | 5 (4.3%)                                               | 1 (7.7%)                                               | 18 (4.0%)         |
| IMC                        | 5 (1.6%)                              | 0 (0%)                                                 | 0 (0%)                                                 | 5 (1.1%)          |
| IPC                        | 5 (1.6%)                              | 3 (2.6%)                                               | 0 (0%)                                                 | 8 (1.8%)          |
| <b>Histological grades</b> |                                       |                                                        |                                                        |                   |
| I                          | 28 (8.7%)                             | 11 (9.6%)                                              | 1 (7.7%)                                               | 40 (8.9%)         |
| II                         | 242 (75.2%)                           | 87 (75.7%)                                             | 12 (92.3%)                                             | 341 (75.8%)       |
| III                        | 52 (16.1%)                            | 17 (14.8%)                                             | 0 (0%)                                                 | 69 (15.3%)        |
| <b>T category</b>          |                                       |                                                        |                                                        |                   |
| T1                         | 127 (39.4%)                           | 55 (47.8%)                                             | 7 (53.8%)                                              | 189 (42.0%)       |
| T2                         | 186 (57.8%)                           | 58 (50.4%)                                             | 5 (38.5%)                                              | 249 (55.3%)       |
| T3                         | 9 (2.8%)                              | 2 (1.7%)                                               | 1 (7.7%)                                               | 12 (2.7%)         |
| <b>N</b>                   |                                       |                                                        |                                                        |                   |
| N0                         | 249 (77.3%)                           | 86 (74.8%)                                             | 7 (53.8%)                                              | 342 (76.0%)       |
| N1                         | 60 (18.6%)                            | 24 (20.9%)                                             | 4 (30.8%)                                              | 88 (19.6%)        |
| N2                         | 7 (2.1%)                              | 3 (2.6%)                                               | 1 (7.7%)                                               | 11 (2.4%)         |
| N3                         | 6 (1.8%)                              | 2 (1.7%)                                               | 1 (7.7%)                                               | 9 (2.0%)          |
| <b>ER</b>                  |                                       |                                                        |                                                        |                   |
| Negative                   | 99 (30.7%)                            | 23 (20.0%)                                             | 1 (7.7%)                                               | 123 (27.3%)       |
| Positive                   | 223 (69.3%)                           | 92 (80.0%)                                             | 12 (92.3%)                                             | 327 (72.7%)       |
| <b>PR</b>                  |                                       |                                                        |                                                        |                   |
| Negative                   | 123 (38.2%)                           | 35 (30.4%)                                             | 1 (7.7%)                                               | 159 (35.3%)       |
| Positive                   | 199 (61.8%)                           | 80 (69.6%)                                             | 12 (92.3%)                                             | 291 (64.7%)       |
| <b>HER-2</b>               |                                       |                                                        |                                                        |                   |
| Negative                   | 217 (67.4%)                           | 93 (80.9%)                                             | 12 (92.3%)                                             | 322 (71.6%)       |
| Positive                   | 105 (32.6%)                           | 22 (19.1%)                                             | 1 (7.7%)                                               | 128 (28.4%)       |
| <b>Ki-67</b>               |                                       |                                                        |                                                        |                   |
| <14                        | 80 (24.8%)                            | 32 (27.8%)                                             | 3 (23.1%)                                              | 115 (25.6%)       |
| ≥14                        | 242 (75.2%)                           | 83 (72.2%)                                             | 10 (76.9%)                                             | 335 (74.4%)       |
| <b>Subtypes</b>            |                                       |                                                        |                                                        |                   |
| HER2+                      | 105 (32.6%)                           | 22 (19.1%)                                             | 1 (7.7%)                                               | 128 (28.4%)       |
| HR + HER2–                 | 169 (52.5%)                           | 81 (70.4%)                                             | 11 (84.6%)                                             | 261 (58.0%)       |
| TNBC                       | 48 (14.9%)                            | 12 (10.4%)                                             | 1 (7.7%)                                               | 61 (13.6%)        |

IDC, invasive ductal carcinoma; ILC, invasive lobular carcinoma; IMC, invasive mucinous carcinoma; IPC, invasive papillary carcinoma; TNBC, triple-negative breast cancer; ER, estrogen receptor; PR, progesterone receptor; HER-2, human epidermal growth factor receptor 2.

Table 7: Baseline characteristics of the cN0 cohort.

subgroups presented superior CCRs, with the T3 subgroup exhibiting the highest sensitivity (Supplementary Tables S12–S14). These results demonstrate that Semi-ALNP maintains strong diagnostic efficacy across diverse clinical and pathological strata.

## Discussion

Currently, ALND is increasingly being replaced by SLNB and radiotherapy, with some patients even

|                  | Pathological examination |                |       | CCR              | Sensitivity     | Specificity      | FPR            | FNR            |
|------------------|--------------------------|----------------|-------|------------------|-----------------|------------------|----------------|----------------|
|                  | Metastasis               | Non-metastasis | Total |                  |                 |                  |                |                |
| Clinical imaging |                          |                |       |                  |                 |                  |                |                |
| Suspicious       | 0                        | 0              | 0     | 76.00% (342/450) | 0% (0/108)      | 100% (342/342)   | 0% (0/342)     | 100% (108/108) |
| Unsuspectious    | 108                      | 342            | 450   |                  |                 |                  |                |                |
| Total            | 108                      | 342            | 450   |                  |                 |                  |                |                |
| Semi-ALNP        |                          |                |       |                  |                 |                  |                |                |
| Suspicious       | 99                       | 34             | 133   | 90.44% (407/450) | 91.67% (99/108) | 90.06% (308/342) | 9.94% (34/342) | 8.33% (9/108)  |
| Unsuspectious    | 9                        | 308            | 317   |                  |                 |                  |                |                |
| Total            | 108                      | 342            | 450   |                  |                 |                  |                |                |

cN0, clinically axillary lymph node-negative; Semi-ALNP, semisupervised MLP multiclassifier; CCR, correct classification rate; FPR, false-positive rate; FNR, false-negative rate.

**Table 8: Comparison of the diagnostic efficacy of clinical imaging evaluation (ultrasound/mammography/MRI) and Semi-ALNP in cN0 patients.**

omitting axillary surgery.<sup>2,10</sup> The SOUND trial revealed that for small BCs with negative axillary ultrasound results, omitting axillary surgery does not compromise treatment outcomes.<sup>42</sup> However, axillary status remains critical for treatment decisions. ALNM is a key indication for NAC and radiotherapy. For HR + HER2- BC, ALNM is an important factor in the clinical decision to use CDK4/6 inhibitors as part of adjuvant intensification therapy, as demonstrated by the MonarchE trial, which highlighted the benefits of adding abemaciclib to endocrine therapy for HR + HER2- BC patients with  $\geq 4$  positive nodes or 1–3 positive nodes with high-risk features.<sup>43</sup> Similarly, the NATALEE trial demonstrated that adding ribociclib to endocrine therapy significantly improved invasive disease-free survival in patients with HR + HER2-early BC with stage II or III disease.<sup>44</sup> For HER2+ BC, the decision to extend treatment with neratinib is often based on ALNM status. Omitting axillary surgery in these patients may lead to understaging and inadequate treatment.<sup>45,46</sup> Therefore, there is a crucial need to develop novel and precise methods for accurate assessment of ALNs.

In this study, a semisupervised MLP classifier was developed to predict ALNM from chest-enhanced CT radiomics in BC patients. Eleven supervised classifiers—including logistic regression, naive Bayes, SVM, k-NN, LDA, QDA, random forest, ExtraTrees, XGBoost, LightGBM, and AdaBoost—were first trained on an “extreme-case” cohort composed of unequivocally non-metastatic (“ultra-low risk”) and fully metastatic (“ultra-high risk”) nodes. Although several models achieved high AUCs ( $>0.90$ ), the MLP outperformed all others, attaining an AUC of 0.959 (95% CI 0.937–0.981), a sensitivity of 89.9% (95% CI 84.7–93.9%), a specificity of 93.2%, and an accuracy of 92.1%. Other methods have specific drawbacks: logistic regression’s linear decision boundary may miss complex nonlinear patterns; naive Bayes relies on a strong feature-independence assumption often violated by correlated radiomic features; k-nearest neighbors show diminished discriminative

power in high-dimensional feature spaces; and tree-based ensembles require extensive hyperparameter tuning to control variance and prevent overfitting.

To mitigate overfitting arising from a small labeled set (597 metastatic nodes vs. 1172 non-metastatic nodes), the MLP model was used to pseudo-label 2422 unlabeled nodes via a Youden index–derived cutoff (0.319), creating a pooled dataset of 4191 nodes for training the final Semi-ALNP. This semisupervised approach broadened the training distribution and maintained robust sensitivity across risk strata (0.883, 0.875, 0.910, and 0.886 for ultra-low, low, high, and ultra-high risk, respectively). Decision curve analysis further confirmed a net clinical benefit across a wide range of thresholds. Finally, to align with routine clinical workflows, the four risk categories were collapsed into a binary scheme - “unsuspectious” (ultra-low + low) vs. “suspectious” (high + ultra-high)—enhancing interpretability and facilitating real-world implementation.

Whether compared with clinical assessment or with other models (SVM and XGBoost), the performance of Semi-ALNP across both multicenter cohorts highlights its strong clinical potential (Table 6, Table 8, Supplementary Tables S16 and S17). First, given its high sensitivity in the axillary biopsy cohort (163/165), it may identify positive cN + patients, potentially eliminating the need for axillary biopsies prior to ALND or NAT. Second, it could identify high-risk individuals within the cN0 cohort. In a multicenter cohort of 450 cN0 patients, the Semi-ALNP successfully identified 22% (99/450) of high-risk cases. Rather than proceeding directly to axillary surgery, these patients could benefit from individualized preoperative treatments, improving their prognosis. Notably, among 9 cN0/pN + missed diagnoses, 2 had micrometastases, likely due to small tumor burdens causing minimal imaging changes. In a dedicated analysis of low-burden disease, Semi-ALNP correctly identified 11 of 18 patients (61.1%) with only micrometastatic ALNs in the internal cohort and 20 of 22 patients (90.9%) with

micrometastasis only in the external cN0 cohort (Supplementary Table S7), suggesting that despite the resolution limits of CT, the model retains value in detecting micrometastases. This may explain why Semi-ALNP performs better in high-risk cases, where metastatic nodes present more distinct radiomic features. Future studies should investigate imaging differences between micrometastatic and macrometastatic ALNs, explore whether SLNB can be safely omitted in patients who are negative by Semi-ALNP, and compare CT-based radiomics with MRI or combined PET-CT to enhance low-burden disease detection.

The Semi-ALNP is the first method to integrate semisupervised learning with multiclass classification for lymph node assessment. Unlike most existing models that predict ALNM based on features derived from the primary tumor or at the patient level, our approach focuses on the lymph node level, enabling more granular and individualized predictions. Built on routine chest-enhanced CT, the model ensures objectivity and reduces reliance on operator-dependent imaging modalities such as ultrasound. While ultrasound and MRI offer advantages in soft-tissue contrast and superficial lesion detection, CT remains a standard and widely accessible modality for preoperative assessment in many regions, with high sensitivity for ALN detection.<sup>47</sup> The Semi-ALNP demonstrated relatively high diagnostic performance across subgroups stratified by age, tumor size, molecular subtype and histologic grade (Supplementary Tables S19–S22). By incorporating unlabeled data, the model enhanced its learning capacity and enabled practical node-level risk stratification to support individualized axillary management and the development of a precise surgical atlas. The performance fluctuations observed in certain external subgroups may be attributed to limited sample sizes. For example, in the biopsy-based external validation cohort, the model showed lower specificity (70%) in TNBC patients while maintaining 100% sensitivity, which is consistent with the clinical superiority of sensitivity in aggressive subtypes with high metastatic potential. Additionally, the modest decrease in specificity may partly reflect the imaging challenges associated with TNBC, where lymph nodes often exhibit less typical radiologic features. These findings highlight the need for larger, more diverse datasets and future refinement through threshold tuning or subtype-aware modeling strategies.

Although a novel semisupervised classifier was developed for predicting ALNMs in BC patients, several limitations must be acknowledged. First, the imperfect correspondence between dissected nodes and those visible on CT, along with the inclusion of potentially erroneous pseudolabels, may have introduced label noise. Nonetheless, the model's robust node- and patient-level performance, along with sensitivity analyses (Supplementary Fig. S6), suggests minimal impact

on overall validity. Future work should incorporate intraoperative node localization and improved imaging–pathology coregistration to refine label accuracy, while also considering entropy-based filtering and patient-level bootstrapping to increase model robustness; Second, despite acquisition at high spatial resolution (1-mm slice thickness) to optimize node visualization, CT may still fail to detect very small lymph nodes (<3 mm), and the manual ROI segmentation approach remains susceptible to subjectivity and reproducibility issues, which could affect the model's accuracy. Future prospective studies will explore ultra-high-resolution CT protocols, advanced image reconstruction and denoising algorithms, and deep learning-based automated segmentation techniques to improve the detection and delineation of small-volume ALNs in clinical workflows; Finally, although Semi-ALNP has been retrospectively validated in multiple cohorts, intercenter heterogeneity remains a concern. A prospective multicenter trial is planned to assess its diagnostic accuracy using blinded Semi-ALNP evaluation and histopathology as a reference, this trial will also test whether Semi-ALNP can safely reduce unnecessary axillary biopsies or ALNDs by accurately triaging patients preoperatively. In addition, Semi-ALNP may help identify otherwise occult node-positive cases among clinically node-negative patients for early intervention and assist in tailoring radiation fields when surgery is limited. These efforts aim to integrate Semi-ALNP into clinical workflows to optimize axillary management with reduced invasiveness and improved precision.

#### Contributors

Limeng Qu and Jinfeng Zhu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing; Xilong Mei: Data curation, Resources, Supervision, Writing – review & editing; Zixi Yi: Data curation, Formal analysis; Na Luo, Songlin Yuan, Xuan Liu, Mingwen Liu, Haiqing Xie, Xiongqiang Hu, Liangrui Pan, Qingchun Liang, Qiongyan Zou, Qin Zhou, Danhua Zhang, Meirong Zhou, Lei Pei, Ke Qian, Qian Long, Qitong Chen, Xi Chen and Jiachi Xu: Data curation, Investigation, Validation; Yanhui Li, Jennifer K. Plichta, Qingyao Shang and Meishuo Ouyang: Conceptualization, Methodology, Supervision; Wenjun Yi: Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing.

All authors read and approved the final manuscript. All authors had full access to the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis as well as the decision to submit for publication.

#### Data sharing statement

The datasets and code used and analyzed during the current study available from the corresponding author upon reasonable request.

#### Declaration of interests

All authors have three patents pending related to the subject matter of this study. These patents are currently under application and do not constitute any commercial or financial conflict of interest. The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential

conflict of interest. Funding sources were not involved in the study design, data collection, analysis and interpretation, writing of the report, or decision to submit the article for publication.

#### Acknowledgements

This study was funded by the Science and Technology Innovation Program of Hunan Province (Grant No. 2021SK2026) and the Innovation Platform and Talent Plan of Hunan Province (2023SK4019). Funding sources were not involved in the study design, data collection, analysis and interpretation, writing of the report, or decision to submit the article for publication.

We would like to express our sincere gratitude to Dr. Linhao Song from the School of Mathematics and Statistics, Central South University, China, for his helpful statistical advice and support during the revision process.

#### Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.eclinm.2025.103311>.

#### References

- Chang JM, Leung JWT, Moy L, Ha SM, Moon WK. Axillary nodal evaluation in breast cancer: state of the art. *Radiology*. 2020;295(3):500–515. <https://doi.org/10.1148/radiol.2020192534>.
- Giuliano AE, Ballman KV, McCall L, et al. Effect of axillary dissection vs No axillary dissection on 10-year overall survival among women with invasive breast cancer and sentinel node metastasis: the ACOSOG Z0011 (alliance) randomized clinical trial. *JAMA*. 2017;318(10):918–926. <https://doi.org/10.1001/jama.2017.11470>.
- Bartels SAL, Donker M, Poncet C, et al. Radiotherapy or surgery of the axilla after a positive sentinel node in breast cancer: 10-year results of the randomized controlled EORTC 10981-22023 AMAROS trial. *J Clin Oncol*. 2023;41(12):2159–2165. <https://doi.org/10.1200/JCO.22.01565>.
- Johnston SRD, Toi M, O'Shaughnessy J, et al. Abemaciclib plus endocrine therapy for hormone receptor-positive, HER2-negative, node-positive, high-risk early breast cancer (monarchE): results from a preplanned interim analysis of a randomised, open-label, phase 3 trial. *Lancet Oncol*. 2023;24(1):77–90. [https://doi.org/10.1016/S1470-2045\(22\)00694-5](https://doi.org/10.1016/S1470-2045(22)00694-5).
- Abe H. Ongoing demand for radiologists in preoperative axillary lymph node assessment. *Radiology*. 2021;300(1):55–56. <https://doi.org/10.1148/radiol.202110369>.
- Bundred NJ, Barnes NL, Rutgers E, Donker M. Is axillary lymph node clearance required in node-positive breast cancer? *Nat Rev Clin Oncol*. 2015;12(1):55–61. <https://doi.org/10.1038/nrclinonc.2014.188>.
- de Boniface J, Appelgren M, Szulkin R, et al. Completion axillary lymph node dissection for the identification of pN2-3 status as an indication for adjuvant CDK4/6 inhibitor treatment: a post-hoc analysis of the randomised, phase 3 SENOMAC trial. *Lancet Oncol*. 2024;25(9):1222–1230. [https://doi.org/10.1016/S1470-2045\(24\)00350-4](https://doi.org/10.1016/S1470-2045(24)00350-4).
- Tinterri C, Canavese G, Gatzemeier W, et al. Sentinel lymph node biopsy versus axillary lymph node dissection in breast cancer patients undergoing mastectomy with one to two metastatic sentinel lymph nodes: sub-analysis of the SINODAR-ONE multicentre randomized clinical trial and reopening of enrolment. *Br J Surg*. 2023;110(9):1143–1152. <https://doi.org/10.1093/bjs/znad215>.
- Reimer T. Omission of axillary sentinel lymph node biopsy in early invasive breast cancer. *Breast*. 2023;67:124–128. <https://doi.org/10.1016/j.breast.2023.01.002>.
- de Boniface J, Filtenborg Tvedskov T, Rydén L, et al. Omitting axillary dissection in breast cancer with sentinel-node metastases. *N Engl J Med*. 2024;390(13):1163–1175. <https://doi.org/10.1056/NEJMoa2313487>.
- Mayerhoefer ME, Materka A, Langs G, et al. Introduction to radiomics. *J Nucl Med*. 2020;61(4):488–495. <https://doi.org/10.2967/jnumed.118.222893>.
- Conti A, Duggento A, Indovina I, Guerrisi M, Toschi N. Radiomics in breast cancer classification and prediction. *Semin Cancer Biol*. 2021;72:238–250. <https://doi.org/10.1016/j.semcancer.2020.04.002>.
- Guiot J, Vaidyanathan A, Deprez L, et al. A review in radiomics: making personalized medicine a reality via routine imaging. *Med Res Rev*. 2022;42(1):426–440. <https://doi.org/10.1002/med.21846>.
- Parekh VS, Jacobs MA. Integrated radiomic framework for breast cancer and tumor biology using advanced machine learning and multiparametric MRI. *NPJ Breast Cancer*. 2017;3:43. <https://doi.org/10.1038/s41523-017-0045-3>.
- Whitney HM, Taylor NS, Drukker K, et al. Additive benefit of radiomics over size alone in the distinction between benign lesions and luminal A cancers on a large clinical breast MRI dataset. *Acad Radiol*. 2019;26(2):202–209. <https://doi.org/10.1016/j.acra.2018.04.019>.
- Li J, Qiao H, Wu F, et al. A novel hypoxia- and lactate metabolism-related signature to predict prognosis and immunotherapy responses for breast cancer by integrating machine learning and bioinformatic analyses. *Front Immunol*. 2022;13:998140. <https://doi.org/10.3389/fimmu.2022.998140>.
- Ozaki J, Fujioka T, Yamaga E, et al. Deep learning method with a convolutional neural network for image classification of normal and metastatic axillary lymph nodes on breast ultrasonography. *Jpn J Radiol*. 2022;40(8):814–822. <https://doi.org/10.1007/s11604-022-01261-6>.
- Zheng X, Yao Z, Huang Y, et al. Deep learning radiomics can predict axillary lymph node status in early-stage breast cancer. *Nat Commun*. 2020;11(1):1236. <https://doi.org/10.1038/s41467-020-15027-z>.
- Bove S, Comes MC, Lorusso V, et al. A ultrasound-based radiomic approach to predict the nodal status in clinically negative breast cancer patients. *Sci Rep*. 2022;12(1):7914. <https://doi.org/10.1038/s41598-022-11876-4>.
- Yu Y, He Z, Ouyang J, et al. Magnetic resonance imaging radiomics predicts preoperative axillary lymph node metastasis to support surgical decisions and is associated with tumor microenvironment in invasive breast cancer: a machine learning, multicenter study. *EBioMedicine*. 2021;69:103460. <https://doi.org/10.1016/j.ebiom.2021.103460>.
- Zhang J, Li L, Zhe X, et al. The diagnostic performance of machine learning-based radiomics of DCE-MRI in predicting axillary lymph node metastasis in breast cancer: a meta-analysis. *Front Oncol*. 2022;12:799209. <https://doi.org/10.3389/fonc.2022.799209>.
- Sun S, Mutasa S, Liu MZ, et al. Deep learning prediction of axillary lymph node status using ultrasound images. *Comput Biol Med*. 2022;143:105250. <https://doi.org/10.1016/j.combiomed.2022.105250>.
- Song B-I. A machine learning-based radiomics model for the prediction of axillary lymph-node metastasis in breast cancer. *Breast Cancer*. 2021;28(3):664–671. <https://doi.org/10.1007/s12282-020-01202-z>.
- Tan H, Gan F, Wu Y, et al. Preoperative prediction of axillary lymph node metastasis in breast carcinoma using radiomics features based on the fat-suppressed T2 sequence. *Acad Radiol*. 2020;27(9):1217–1225. <https://doi.org/10.1016/j.acra.2019.11.004>.
- Han L, Zhu Y, Liu Z, et al. Radiomic nomogram for prediction of axillary lymph node metastasis in breast cancer. *Eur Radiol*. 2019;29(7):3820–3829. <https://doi.org/10.1007/s00330-018-5981-2>.
- Zhou L-Q, Wu X-L, Huang S-Y, et al. Lymph node metastasis prediction from primary breast cancer US images using deep learning. *Radiology*. 2020;294(1):19–28. <https://doi.org/10.1148/radiol.2019190372>.
- Chen S-T, Lai H-W, Chang JH-M, et al. Diagnostic accuracy of preoperative breast magnetic resonance imaging (MRI) in predicting axillary lymph node metastasis: variations in intrinsic subtypes, and strategy to improve negative predictive value-an analysis of 2473 invasive breast cancer patients. *Breast Cancer*. 2023;30(6):976–985. <https://doi.org/10.1007/s12282-023-01488-9>.
- Zhu Y, Ma Y, Zhai Z, et al. Radiomics in cone-beam breast CT for the prediction of axillary lymph node metastasis in breast cancer: a multi-center multi-device study. *Eur Radiol*. 2024;34(4):2576–2589. <https://doi.org/10.1007/s00330-023-10256-4>.
- Tang Y, Che X, Wang W, Su S, Nie Y, Yang C. Radiomics model based on features of axillary lymphatic nodes to predict axillary lymphatic node metastasis in breast cancer. *Med Phys*. 2022;49(12):7555–7566. <https://doi.org/10.1002/mp.15873>.
- Urso L, Manco L, Castello A, et al. PET-derived radiomics and artificial intelligence in breast cancer: a systematic review. *Int J Mol Sci*. 2022;23(21). <https://doi.org/10.3390/ijms232113409>.
- Song SE, Woo OH, Cho Y, Cho KR, Park KH, Kim JW. Prediction of axillary lymph node metastasis in early-stage triple-negative

- breast cancer using multiparametric and radiomic features of breast MRI. *Acad Radiol*. 2023;30(Suppl 2):S25–S37. <https://doi.org/10.1016/j.acra.2023.05.025>.
- 32 Qu L, Mei X, Yi Z, et al. An unsupervised learning model based on CT radiomics features accurately predicts axillary lymph node metastasis in breast cancer patients-diagnostic study. *Int J Surg*. 2024. <https://doi.org/10.1097/j.s9.0000000000001778>.
  - 33 Chen Y, Liu Y, Wang C, et al. BRAIxDet: learning to detect malignant breast lesion with incomplete annotations. *Med Image Anal*. 2024;96:103192. <https://doi.org/10.1016/j.media.2024.103192>.
  - 34 Fan Z, Gong P, Tang S, et al. Joint localization and classification of breast masses on ultrasound images using an auxiliary attention-based framework. *Med Image Anal*. 2023;90:102960. <https://doi.org/10.1016/j.media.2023.102960>.
  - 35 Huynh T, Nibali A, He Z. Semi-supervised learning for medical image classification using imbalanced training data. *Comput Methods Programs Biomed*. 2022;216:106628. <https://doi.org/10.1016/j.cmpb.2022.106628>.
  - 36 Zhang J, Cui Z, Zhou L, et al. Breast fibroglandular tissue segmentation for automated BPE quantification with iterative cycle-consistent semi-supervised learning. *IEEE Trans Med Imaging*. 2023;42(12):3944–3955. <https://doi.org/10.1109/TMI.2023.3319646>.
  - 37 Besson FL, Henry T, Meyer C, et al. Rapid contour-based segmentation for 18F-FDG PET imaging of lung tumors by using ITK-SNAP: comparison to expert-based segmentation. *Radiology*. 2018;288(1):277–284. <https://doi.org/10.1148/radiol.2018171756>.
  - 38 van Griethuysen JJM, Fedorov A, Parmar C, et al. Computational radiomics system to decode the radiographic phenotype. *Cancer Res*. 2017;77(21):e104–e107. <https://doi.org/10.1158/0008-5472.CAN-17-0339>.
  - 39 Pedregosa F, Varoquaux G, Gramfort A, et al. Scikit-learn: machine learning in Python. *J Mach Learn Res*. 2011;12(null):2825–2830.
  - 40 Huang Y, Zhu T, Zhang X, et al. Longitudinal MRI-based fusion novel model predicts pathological complete response in breast cancer treated with neoadjuvant chemotherapy: a multicenter, retrospective study. *eClinicalMedicine*. 2023;58:101899. <https://doi.org/10.1016/j.eclinm.2023.101899>.
  - 41 Dessau RB, Pipper CB. ["R"-project for statistical computing]. *Ugeskr Laeger*. 2008;170(5):328–330.
  - 42 Gentilini OD, Botteri E, Sangalli C, et al. Sentinel lymph node biopsy vs No axillary surgery in patients with small breast cancer and negative results on ultrasonography of axillary lymph nodes: the SOUND randomized clinical trial. *JAMA Oncol*. 2023;9(11):1557–1564. <https://doi.org/10.1001/jamaoncol.2023.3759>.
  - 43 Johnston SRD, Harbeck N, Hegg R, et al. Abemaciclib combined with endocrine therapy for the adjuvant treatment of HR+, HER2-, node-positive, high-risk, early breast cancer (monarchE). *J Clin Oncol*. 2020;38(34):3987–3998. <https://doi.org/10.1200/JCO.20.02514>.
  - 44 Slamon D, Lipatov O, Nowecki Z, et al. Ribociclib plus endocrine therapy in early breast cancer. *N Engl J Med*. 2024;390(12):1080–1091. <https://doi.org/10.1056/NEJMoa2305488>.
  - 45 Martin M, Holmes FA, Ejlertsen B, et al. Neratinib after trastuzumab-based adjuvant therapy in HER2-positive breast cancer (ExteNET): 5-year analysis of a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Oncol*. 2017;18(12):1688–1700. [https://doi.org/10.1016/S1470-2045\(17\)30717-9](https://doi.org/10.1016/S1470-2045(17)30717-9).
  - 46 Chan A, Delaloge S, Holmes FA, et al. Neratinib after trastuzumab-based adjuvant therapy in patients with HER2-positive breast cancer (ExteNET): a multicentre, randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Oncol*. 2016;17(3):367–377. [https://doi.org/10.1016/S1470-2045\(15\)00551-3](https://doi.org/10.1016/S1470-2045(15)00551-3).
  - 47 Groheux D. Breast cancer systemic staging (comparison of computed tomography, bone scan, and 18F-fluorodeoxyglucose PET/computed tomography). *PET Clin*. 2023;18(4):503–515. <https://doi.org/10.1016/j.cpet.2023.04.006>.